



Yakeen Plus (2025)

Practice Test - 20

DURATION : 200 Minutes

DATE : 02/01/2025

M. MARKS : 720

Topics Covered

Physics :	Electrostatics, Current Electricity, Magnetic Effects of Current and Magnetism, Electromagnetic Induction and Alternating Currents, Electromagnetic Waves
Chemistry :	Thermodynamics and Thermochemistry, Chemical Kinetics, Alcohols, Ethers and Phenols, Aldehydes, Ketones and Carboxylic Acids, The p-Block Elements (Group 13,14,15, 16, 17, 18), The d and f-Block Elements
Biology :	(Botany) : Sexual Reproduction in Flowering Plant, Principle of Inheritance and Variation, Microbes in Human Welfare (Zoology) : Human Reproduction, Reproductive Health, Evolution

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are four Section in the Question Paper, Section I, II, III & IV consisting of Section-I (**Physics**), Section-II (**Chemistry**), Section-III (**Botany**) & Section IV (**Zoology**) and having **50 Questions** in each Subject and each subject is divided in two Section, **Section A** consisting 35 questions (all questions are compulsory) and **Section B** consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

Answer Key

Q1 (1)
Q2 (2)
Q3 (3)
Q4 (3)
Q5 (2)
Q6 (2)
Q7 (1)
Q8 (1)
Q9 (3)
Q10 (4)
Q11 (4)
Q12 (2)
Q13 (2)
Q14 (2)
Q15 (1)
Q16 (1)
Q17 (2)
Q18 (4)
Q19 (1)
Q20 (3)
Q21 (3)
Q22 (1)
Q23 (2)
Q24 (2)
Q25 (4)
Q26 (1)
Q27 (1)

Q28 (4)
Q29 (1)
Q30 (4)
Q31 (1)
Q32 (1)
Q33 (3)
Q34 (3)
Q35 (3)
Q36 (3)
Q37 (1)
Q38 (1)
Q39 (4)
Q40 (2)
Q41 (2)
Q42 (3)
Q43 (1)
Q44 (3)
Q45 (4)
Q46 (4)
Q47 (1)
Q48 (3)
Q49 (3)
Q50 (1)
Q51 (4)
Q52 (4)
Q53 (4)
Q54 (3)



Q55 (3)
Q56 (3)
Q57 (1)
Q58 (1)
Q59 (2)
Q60 (2)
Q61 (3)
Q62 (2)
Q63 (3)
Q64 (3)
Q65 (1)
Q66 (1)
Q67 (2)
Q68 (1)
Q69 (1)
Q70 (2)
Q71 (2)
Q72 (1)
Q73 (3)
Q74 (3)
Q75 (4)
Q76 (4)
Q77 (3)
Q78 (1)
Q79 (4)
Q80 (3)
Q81 (2)
Q82 (3)
Q83 (4)

Q84 (1)
Q85 (2)
Q86 (1)
Q87 (2)
Q88 (4)
Q89 (3)
Q90 (1)
Q91 (2)
Q92 (2)
Q93 (2)
Q94 (2)
Q95 (1)
Q96 (4)
Q97 (3)
Q98 (4)
Q99 (4)
Q100 (1)
Q101 (2)
Q102 (4)
Q103 (2)
Q104 (4)
Q105 (3)
Q106 (4)
Q107 (4)
Q108 (4)
Q109 (3)
Q110 (2)
Q111 (3)
Q112 (3)



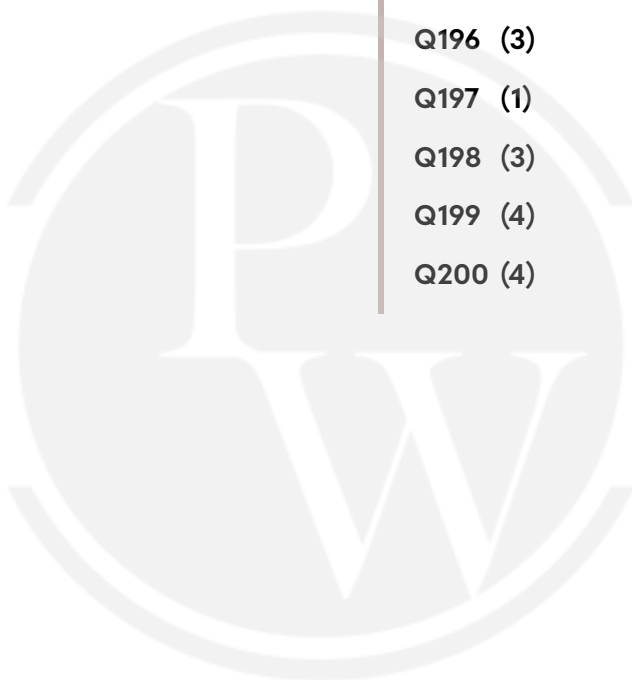
Q113 (4)
Q114 (4)
Q115 (4)
Q116 (4)
Q117 (4)
Q118 (3)
Q119 (3)
Q120 (3)
Q121 (4)
Q122 (3)
Q123 (2)
Q124 (3)
Q125 (1)
Q126 (4)
Q127 (1)
Q128 (4)
Q129 (2)
Q130 (4)
Q131 (2)
Q132 (3)
Q133 (1)
Q134 (4)
Q135 (3)
Q136 (4)
Q137 (3)
Q138 (4)
Q139 (4)
Q140 (1)
Q141 (3)

Q142 (3)
Q143 (4)
Q144 (1)
Q145 (1)
Q146 (3)
Q147 (2)
Q148 (2)
Q149 (4)
Q150 (1)
Q151 (1)
Q152 (3)
Q153 (3)
Q154 (4)
Q155 (3)
Q156 (1)
Q157 (3)
Q158 (2)
Q159 (3)
Q160 (2)
Q161 (4)
Q162 (4)
Q163 (1)
Q164 (2)
Q165 (4)
Q166 (3)
Q167 (4)
Q168 (2)
Q169 (1)
Q170 (3)



Q171 (2)
Q172 (4)
Q173 (3)
Q174 (3)
Q175 (3)
Q176 (3)
Q177 (4)
Q178 (1)
Q179 (3)
Q180 (3)
Q181 (3)
Q182 (3)
Q183 (4)
Q184 (1)
Q185 (4)

Q186 (3)
Q187 (4)
Q188 (4)
Q189 (2)
Q190 (1)
Q191 (1)
Q192 (1)
Q193 (1)
Q194 (3)
Q195 (2)
Q196 (3)
Q197 (1)
Q198 (3)
Q199 (4)
Q200 (4)



Hints & Solutions

Q1 Text Solution:

(1)

Charge Q is 0 to 5C steadily

Potential difference $V = \frac{Q}{C}$

$$V_{\max} = \frac{5}{2 \times 10^{-6}} \Rightarrow V_{\max} = 2.5 \times 10^6 \text{ Volt}$$

$V \propto Q$

{ Q is steadily so V is also steadily}

Q2 Text Solution:

(2)

$$R = \frac{mv}{qB}$$

$$R = \frac{\sqrt{2mk}}{qB} \{k = \text{same}, B = \text{same}\}$$

$$\frac{q_1}{q_2} = \left(\sqrt{\frac{m_1}{m_2}} \right) \left(\frac{R_2}{R_1} \right)$$

$$\frac{q_1}{q_2} = \left(\sqrt{\frac{9}{4}} \right) \left(\frac{5}{6} \right)$$

$$\frac{q_1}{q_2} = \frac{5}{4}$$

Q3 Text Solution:

(3)

$$\text{Resonant frequency } F_r = \frac{1}{2\pi\sqrt{LC}}$$

By adding a capacitor in series

equivalent capacitance decreases

Hence resonant frequency increases.

Q4 Text Solution:

(3)

Magnetic field at centre

$$B = 4 \left(\frac{\mu_0 I}{4\pi \left(\frac{L}{2} \right)} \right) (2\sin 45^\circ)$$

$$B = 2\sqrt{2} \frac{\mu_0 I}{\pi L}$$

Magnetic flux in small loop

$$\phi = \beta I^2$$

$$\phi = 2\sqrt{2} \frac{\mu_0 I}{\pi L} I^2$$

$$\text{Mutual Inductance } M = \frac{\phi_s}{I_p}$$

$$M = 2\sqrt{2} \frac{\mu_0 I^2}{\pi L}$$

Q5 Text Solution:

(2)

$$I_{rms} = \frac{V_{rms}}{x_c} \left\{ x_c = \frac{1}{\omega c} \right\}$$

$$I_{rms} = V_{rms} \omega c$$

$$C = \frac{I_{rms}}{\omega V_{rms}} \Rightarrow C = \frac{6.9 \times 10^{-6}}{600 \times 230}$$

$$C = 50 \text{ pF}$$

Q6 Text Solution:

(2)

$$B_1 = \frac{2\mu_0 i}{2R_1} \quad 2 \times 2\pi R_1 = 5 \times 2\pi R_2$$

$$B_2 = \frac{5\mu_0 i}{2R_2} \frac{R_1}{R_2} = \frac{5}{2}$$

$$\frac{B_2}{B_1} = \frac{5}{2} \frac{R_1}{R_2}$$

$$= \frac{5}{2} \times \frac{5}{2} = \frac{25}{4}$$

Q7 Text Solution:

(1)

Magnetic moment = pole strength $\times$ length

Q8 Text Solution:

(1)

$$\frac{E_0}{B_0} = C$$

$$B_0 = \frac{E_0}{C} = \frac{540}{3 \times 10^8} = 18 \times 10^{-7}$$

Q9 Text Solution:

(3)

$$f = \frac{qB}{2\pi m} = \frac{1.6 \times 10^{-19} \times 1 \times 10^{-4}}{2 \times 3.14 \times 9 \times 10^{-31}}$$

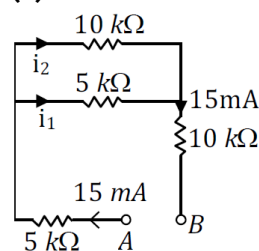
$$= 0.028 \times 10^{-23+31}$$

$$= 0.028 \times 10^8$$

$$= 2.8 \times 10^6 \text{ Hz}$$

Q10 Text Solution:

(4)



$$\frac{i_1}{i_2} = \frac{R_2}{R_1} = \frac{10}{5} = \frac{2}{1}$$

$$i_1 = \frac{2}{3} \times 15 \text{ mA} = 10 \text{ mA}$$

$$V_A - 15 \times 5 - 10 \times 5 - 15 \times 10 = V_B$$

$$V_A - V_B = 275 \text{ V}$$

Q11 Text Solution:

(4)

$$H = i^2 R t$$

$$\frac{\Delta H}{H} = \frac{2\Delta i}{i} + \frac{\Delta R}{R} + \frac{\Delta t}{t}$$

$$= 2 \times 2\% + 1\% + 3\% = 8\%$$

Q12 Text Solution:

(2)

$$P_1 = \cos \phi_1 = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + R^2}} = \frac{R}{R\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$P_2 = \cos \phi_2 = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (R-R)^2}} = \frac{R}{R} = 1$$

$$\frac{P_1}{P_2} = \frac{\frac{1}{\sqrt{2}}}{1} = \frac{1}{\sqrt{2}}$$

Q13 Text Solution:

(2)

$$\vec{B} \cdot \vec{a} = 0, \left[\vec{B} \perp \vec{a} \right]$$

$$2\alpha - 12 = 0$$

$$\alpha = 6$$

Q14 Text Solution:

(2)

$$\frac{B_x}{B_y} = \frac{\frac{\mu_0 N_1 I}{2r}}{\frac{\mu_0 N_2 I}{2r}} = \frac{N_1}{N_2} = \frac{200}{400} = \frac{1}{2}$$

Q15 Text Solution:

(1)

$$\text{By WSB, } R_{eq} = \frac{8 \times 8}{8+8} = 4\Omega$$

$$I = \frac{V}{R_{eq}} = \frac{40}{4} = 10 \text{ A}$$

Q16 Text Solution:

(1)

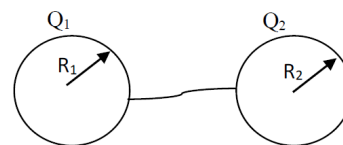
$$C_{eq} = C_1 + C_2 + C_3 + C_4 = 1 + 2 + 3 + 4 = 10 \mu\text{F}$$

$$q = C_{eq} V = 10 \mu\text{F} \times 20 = 200 \mu\text{C}$$

Q17 Text Solution:

(2)

At equilibrium, their potential will be same as they are connected by a conducting wire.



Given: $R_1 = 5 \text{ mm}$ and $R_2 = 10 \text{ mm}$.

Let at equilibrium, charges on both the spheres be Q_1 and Q_2 respectively. Then,

$$V_1 = V_2$$

$$\Rightarrow \frac{KQ_1}{R_1} = \frac{KQ_2}{R_2}$$

$$\Rightarrow \frac{Q_1}{R_1} = \frac{Q_2}{R_2}$$

$$\Rightarrow \frac{Q_1}{Q_2} = \frac{R_1}{R_2} = \frac{5}{10} = \frac{1}{2}$$

Now, electric field on surface of spherical conductor is given by $E = \frac{KQ}{R^2}$.

$$\therefore \frac{E_1}{E_2} = \frac{\frac{KQ_1}{R_1^2}}{\frac{KQ_2}{R_2^2}} = \frac{Q_1}{Q_2} \times \frac{R_2^2}{R_1^2}$$

$$\Rightarrow \frac{E_1}{E_2} = \frac{1}{2} \times \left(\frac{10}{5}\right)^2 = 2$$

$$\therefore \frac{E_1}{E_2} = \frac{2}{1}$$

Q18 Text Solution:

(4)

From the wave equation, we get $v = \frac{\omega}{k} = \frac{4 \times 10^8}{5}$

Now, amplitude of electric field is given by

$$E_0 = vB_0$$

$$\therefore E_0 = \frac{4 \times 10^8}{5} \times 5 \times 10^{-6} = 4 \times 10^2 \text{ Vm}^{-1}$$

Q19 Text Solution:

(1)

Given circuit is a balance Wheatstone bridge.

So, there will be no current in 5Ω resistance.

$$R_1 = \frac{6 \times 12}{6+12} = 4\Omega$$

$$R_{eq} = 4 + 2 = 6\Omega$$

$$\therefore I = \frac{V}{R_{eq}} = \frac{6}{6} = 1 \text{ A}$$

Q20 Text Solution:

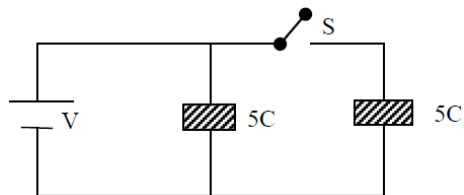
In case I, when the switch was closed and dielectric was not inserted:

$$E_1 = \frac{1}{2} (2C)V^2 = V^2$$





In case II, when the switch was open and dielectric was inserted:



$$E_2 = \frac{1}{2} (5C)V^2 + \frac{1}{2} \frac{Q^2}{5C}$$

$$\Rightarrow E_2 = \frac{1}{2} (5C)V^2 + \frac{1}{2} \frac{(CV)^2}{5C}$$

$$\Rightarrow E_2 = \frac{1}{2} (5C)V^2 + \frac{1}{10} CV^2$$

$$\Rightarrow E_2 = \left(\frac{5}{2} + \frac{1}{10}\right) CV^2 = \frac{26}{10} CV^2$$

$$\therefore E_2 = \frac{13}{5} CV^2$$

Now, $\frac{E_1}{E_2} = \frac{5}{13}$

Q21 Text Solution:

(3)

$$\frac{\mu_0 i R^2}{2(R^2 + x^2)^{3/2}} = \frac{1}{2\sqrt{2}} \frac{\mu_0 i}{2R}$$

$$2\sqrt{2}R^3 = (R^2 + x^2)^{3/2}$$

$$(2\sqrt{2})^{2/3} R^2 = R^2 + x^2$$

$$x^2 = 2R^2 - R^2 = R^2 \Rightarrow x = R$$

Q22 Text Solution:

(1)

According to Faraday's law of induction

$$\text{Induced e.m.f. } \varepsilon = -\frac{d\phi}{dt} = -(100t)$$

Induced current i at $t = 2$ sec.

$$= \left| \frac{\varepsilon}{R} \right| = +\frac{100 \times 2}{400} = +0.5 \text{ Amp}$$

Q23 Text Solution:

(2)

By Theory

Q24 Text Solution:

(2)

The charge through the coil = area of current-time ($i-t$) graph

$$q = \frac{1}{2} \times 0.1 \times 4 = 0.2 \text{ C}$$

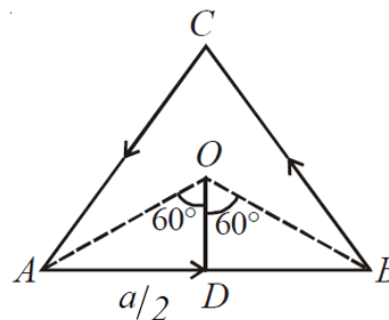
$$q = \frac{\Delta\phi}{R} \quad \therefore \text{Change in flux } (\Delta\phi) = q \times R$$

$$q = 0.2 = \frac{\Delta\phi}{10}$$

$$\Delta\phi = 2 \text{ weber}$$

Q25 Text Solution:

(4)



O is centroid and using the $\triangle OAD$ distance

$$OD = \frac{a}{2\sqrt{3}}$$

By all the three sides AB, BC and CA, direction of magnetic field produced will be same and outward to the plane of paper,

So

B_{total}

$$= 3 \left[\frac{\mu_0 i}{4\pi \left(\frac{a}{2\sqrt{3}}\right)} \left(\sin 60^\circ + \sin 60^\circ \right) \right]$$

$$= \frac{9\mu_0 i}{2\pi a}$$

Q26 Text Solution:

(1)



$$C_{eq} = \frac{C(KC)}{C+KC} = \frac{KC}{K+1}$$

$$24 = \frac{K40}{K+1}$$

$$[K = 1.5]$$

Q27 Text Solution:

(1)

$$R_1 = \rho \frac{L_1}{A_1}$$

$$R_2 = \rho \left(\frac{3L_1}{A_1/3} \right) = 9\rho \frac{L_1}{A_1}$$

$$\therefore \frac{R_2}{R_1} = 9$$



Q28 Text Solution:

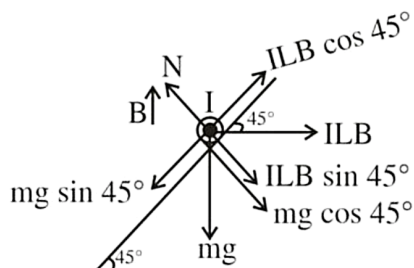
(4)

$$i = \left(\frac{K}{NAB} \right) \theta$$

$$\therefore \frac{d\theta}{di} = \frac{NAB}{K}$$

Q29 Text Solution:

(1)



$$mg \sin 45^\circ = ILB \cos 45^\circ$$

$$\therefore I = \left(\frac{m}{L} \right) \frac{g}{B}$$

$$= \frac{(0.45)(10)}{0.15} = 30 \text{ A}$$

Q30 Text Solution:

(4)

$$v = i_0 X_L = i_0 (\omega L)$$

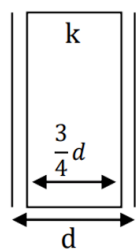
$$= (5)(49\pi)(30 \times 10^{-3}) = 23.1$$

Voltage will lead current by 90° .

$$\therefore v = 23.1 \sin(49\pi t + 60^\circ)$$

Q31 Text Solution:

(1)



$$C_0 = \frac{\epsilon_0 A}{d}$$

$$C = \frac{\epsilon_0 A}{\frac{d}{4} + \frac{3d}{4k}}$$

$$= \frac{4k\epsilon_0 A}{(K+3)d}$$

$$C = \frac{4k}{3+k} C_0$$

Q32 Text Solution:

(1)

If velocity of the charge particle is parallel to magnetic field then particle will not experience any force.

Magnetic force is perpendicular to velocity so speed does not change. Therefore kinetic energy does not change.

Q33 Text Solution:

(3)

For diamagnetic substance $\chi_m < 0$ and $\mu_r < 1$

Q34 Text Solution:

(3)

Velocity has two components, one along magnetic field and other perpendicular to magnetic field, due to which helical motion will take place.

Q35 Text Solution:

(3)

magnetic field at center of loop

$$B_1 = \frac{\mu_0 i}{2R}$$

Magnetic field at $x = \sqrt{3}R$

$$B_2 = \frac{\mu_0 i R^2}{2(R^2 + x^2)^{3/2}} = \frac{\mu_0 i R^2}{2(R^2 + 3R^2)^{3/2}}$$

$$= \frac{\mu_0 i R^2}{2(4R^2)^{3/2}} = \frac{\mu_0 i}{16R}$$

$$\text{So, } \frac{B_1}{B_2} = \frac{16}{2} = \frac{8}{1}$$

$$B_1 : B_2 = 8 : 1$$

Q36 Text Solution:

(3)

$$B = \sqrt{\left(\frac{\mu_0 I}{2R} \right)^2 + \left(\frac{\mu_0 (2I)}{2R} \right)^2}$$

$$= \frac{\mu_0 I}{2R} \sqrt{1 + 4} = \frac{\sqrt{5} \mu_0 I}{2R}$$

Q37 Text Solution:

(1)

$$\chi = \frac{c}{T}, \chi \text{ is susceptibility}$$

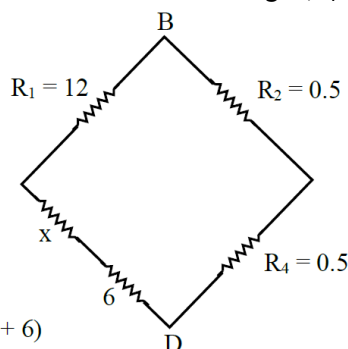
$$\chi = m \left(\frac{1}{T} \right)$$

$$\tan \theta = m = \frac{0.4}{7 \times 10^{-3}} = 57 \text{ K}$$

Q38 Text Solution:

(1)

Potential difference between B & D will be zero in a balanced Wheatstone bridge ($R_1 R_4 = R_2 R_3$)



$$R_3 = (x + 6)$$

$$(12)(0.5) = (0.5)(x + 6)$$

$$x = 6$$

Q39 Text Solution:

(4)

$$P = 110 \text{ W}$$

$$V = 220 \text{ volt}$$

$$P = VI$$

$$110 = 220I$$

$$I = 0.5 \text{ A}$$

$$I = \frac{q}{t} = \frac{ne}{t}$$

$$0.5 = \left(\frac{n}{t}\right) \times 1.6 \times 10^{-19}$$

$$\left(\frac{n}{t}\right) = \frac{0.5}{1.6 \times 10^{-19}}$$

$$= 31.25 \times 10^{17}$$

Q40 Text Solution:

(2)

Conceptual

Q41 Text Solution:

(2)

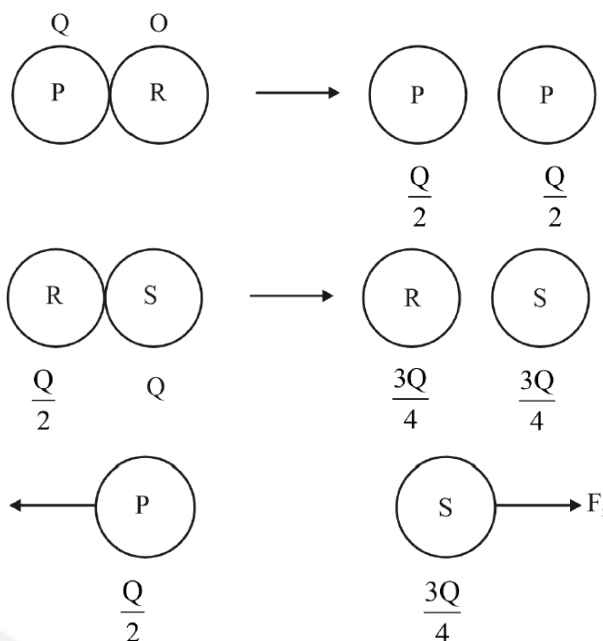
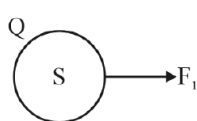
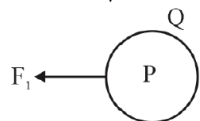
$$\int \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{inside}} = \mu_0 (4i - i) = 3\mu_0 i$$

Q42 Text Solution:

(3)

$$F_1 = \frac{KQ^2}{r^2} = 16$$

...(1)



$$F_2 = \frac{K \frac{Q}{2} \cdot \frac{3Q}{4}}{r^2} = \frac{3}{8} \cdot \frac{KQ^2}{r^2} = \frac{3}{8} \times 16$$

$$F_2 = 6 \text{ N}$$

Q43 Text Solution:

(1)

Conceptual

Q44 Text Solution:

(3)

As dielectric is placed, capacitance will increase.

$$X_C = \frac{1}{C_w}$$

So capacitive Reactance decreases. So impedance of circuit decreases. Hence current increases. So power increases. So glow of the bulb increases.

Q45 Text Solution:

(4)

All capacitors are in parallel.

$$C_{\text{eq}} = C_1 + C_2 + C_3$$

$$= \frac{\epsilon_0 A}{3 \times d} + \frac{\epsilon_0 A}{3 \times 2d} + \frac{\epsilon_0 A}{3 \times 3d}$$

$$= \frac{6\epsilon_0 A + 3\epsilon_0 A + 2\epsilon_0 A}{18d} = \frac{11\epsilon_0 A}{18d}$$

Q46 Text Solution:

(4)

$$\therefore B = 1 \text{ T}$$

At $t = 10$ sec. distance moved by loop;



$$d = v \times t = 2 \times 10 = 20 \text{ cm}$$

i.e. loop completely inside the field

Now, emf induced

$$E = \frac{d\phi}{dt} = B \frac{dA}{dt} = 0$$

(No change in flux)

Q47 Text Solution:

(1)

Resistance

$$R = \frac{V^2}{P} = \frac{(200)^2}{50} = 800 \Omega$$

$$\text{Power dissipated} = \frac{V_1^2}{R}$$

$$= \frac{(100)^2}{800} = \frac{100}{8} = 12.5 \text{ W}$$

Q48 Text Solution:

(3)

In purely capacitive, current leads voltage by 90°

In purely inductive, current lags voltage by 90°

At resonance, $X_C = X_L$

i.e., $Z = R$

Q49 Text Solution:

(3)

$$U = -\vec{M} \cdot \vec{B}$$

$$\Delta U = 2MB = 2 \times 0.5 \times 8 \times 10^{-2}$$

$$= 8 \times 10^{-2} \text{ J}$$

Q50 Text Solution:

(1)

$$\vec{E} \times \vec{B} = \hat{k} \times (2\hat{i} - 2\hat{j})$$

$$= 2\hat{k} \times \hat{i} - 2\hat{k} \times \hat{j}$$

$$= 2\hat{j} + 2\hat{i}$$

Unit vector along

$$\vec{E} \times \vec{B} = \frac{1}{2\sqrt{2}} (2\hat{i} + 2\hat{j})$$

$$= \frac{1}{\sqrt{2}} (\hat{i} + \hat{j})$$

$$C = \frac{1}{\sqrt{2}} (\hat{i} + \hat{j})$$

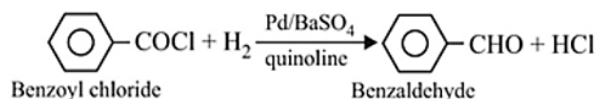
Q51 Text Solution:

(4)

All the given statements regarding $K_2Cr_2O_7$ are correct.

Q52 Text Solution:

(4)



Q53 Text Solution:

(4)

The outer electronic configuration of chromium is $3d^5 4s^1$. It exhibits oxidation states +2, +3, +6. Its +3 oxidation state is highly stable in acidic solution but it gets oxidised readily to +6 oxidation state in alkali solution. Chromium (II) compounds are reducing agents as they are readily converted into chromium (III) compounds even by the oxygen from the air.

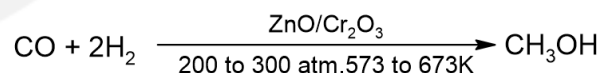
Q54 Text Solution:

(3)

Fluorine and chlorine having low van der Waals forces of attraction are gases. Br_2 is liquid, while iodine with high molecular mass and more van der Waals force of attraction between the molecules is solid at room temperature.

Q55 Text Solution:

(3)



Q56 Text Solution:

(3)

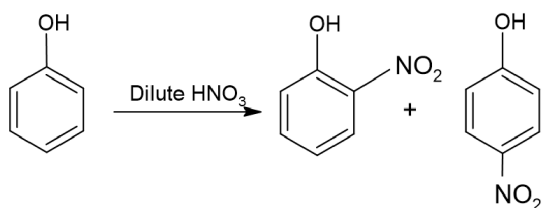
Decrease in activation energy = $58 - 43 = 15$ kJ/mol

Catalyst equally effects the activation energy for both forward and backward reactions.

Q57 Text Solution:

(1)

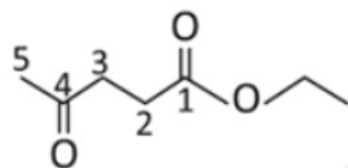




mixture of o-nitrophenol and p-nitrophenol formed.

Q58 Text Solution:

(1)



Ethyl-4-oxopentanoate

Q59 Text Solution:

(2)

Stronger is the acid, smaller will be its pK_a value.
Out of the given formic acid is the strongest
hence it has smallest pK_a value.

Q60 Text Solution:

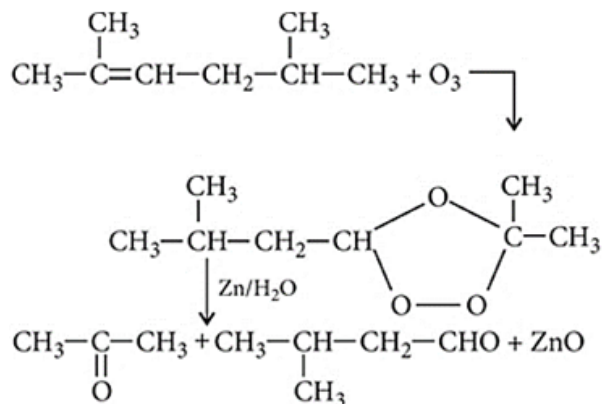
(2)

$\text{Eu}(Z=63) \rightarrow [\text{Xe}]4f^7 6s^2$, $\text{Eu}^{2+} \rightarrow [\text{Xe}]4f^7$:

Due to highly stable half filled configuration Eu^{2+} is stable.

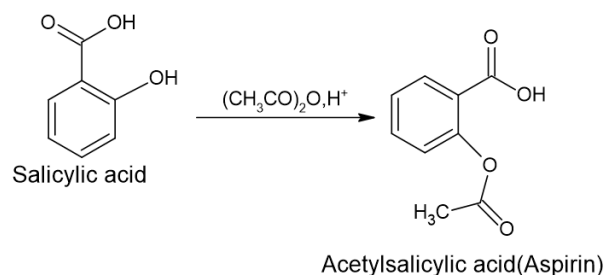
Q61 Text Solution:

(3)



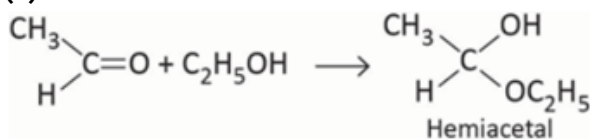
Q62 Text Solution:

(2)



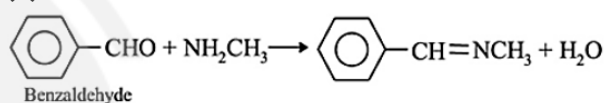
Q63 Text Solution:

(3)



Q64 Text Solution:

(3)



Q65 Text Solution:

(1)

Three $-\text{NO}_2$ groups in picric acid deactivate the phenolate ion. Due to electron withdrawing nature of $-\text{NO}_2$ groups picric acid becomes stronger acid than phenol.

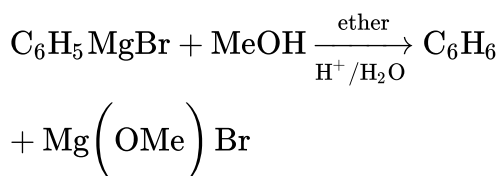
Q66 Text Solution:

(1)

If $\mu = \sqrt{24} = \sqrt{4(4+2)}$ B.M
thus, number of unpaired $e^- = 4$
 $\therefore$ Fe must have +2 charge.

Q67 Text Solution:

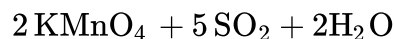
(2)



Q68 Text Solution:

(1)

SO₂ readily decolourises pink violet colour of acidified KMnO₄ solution.



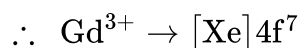
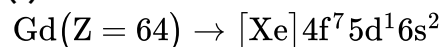
(Pink violet)



(Colourless)

Q69 Text Solution:

(1)

**Q70 Text Solution:**

(2)

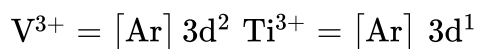
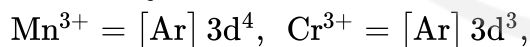
Phenol and ethanol can be distinguished by Iodoform test.

Q71 Text Solution:

(2)

Out of the given species, Cr³⁺ has the highest negative reduction potential. Hence, it cannot be reduced to Cr²⁺ and therefore, is the most stable ion in aqueous solution.

Alternatively,



In Cr³⁺, all three d electrons enter into lower energy t_{2g} orbitals. The lowering of energy is maximum and hence stability is maximum.

Q72 Text Solution:

(1)

In an exothermic reaction, there is the decrease in enthalpy in passing from reactants to products.

Q73 Text Solution:

(3)

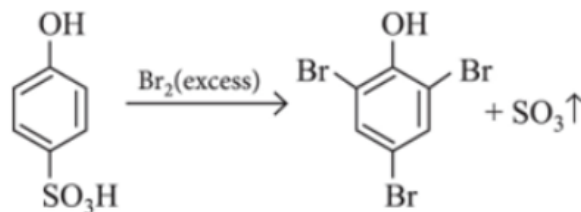
For a first order reaction

$$k = \frac{2.303}{t} \log \frac{[\text{R}]_0}{[\text{R}]} \Rightarrow t = \frac{2.303}{4.606 \times 10^{-3} \text{s}^{-1}}$$

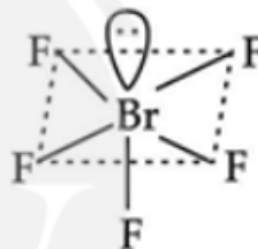
$$\log \left(\frac{2}{0.2} \right) = \frac{2.303 \times 10^3}{4.606} = 500 \text{ s}$$

Q74 Text Solution:

(3)

**Q75 Text Solution:**

(4)



sp³d²-hybridisation

Square pyramidal-shape

Q76 Text Solution:

(4)

Both groups are ring activating groups. So, all o- and p-positions are substituted.

Q77 Text Solution:

(3)

Melting point of group 13 elements decreases down the group but the trend is not regular.

Melting point decreases sharply on moving down the group from B to Ga and then increases from Ga to Tl. Boron has highest melting point (2453 K) because of its very strong crystal lattice.



Gallium has lowest melting point (303 K) and therefore can exist in liquid state during summer.

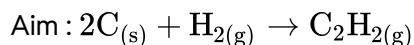
Q78 Text Solution:

(1)

Both white and red phosphorus are not soluble in CS_2 . Only white P is soluble in CS_2 .

Q79 Text Solution:

(4)



Operate (i) + (ii) – (iii), we get

$$\Delta H = -787 + (-286) - (-1301) = \text{mol}^{-1} + 228 \text{ kJ}$$

Q80 Text Solution:

(3)

$$\text{Rate} = -\frac{1}{4} \frac{d[\text{KClO}_3]}{dt} = \frac{1}{3} \frac{d[\text{KClO}_4]}{dt} = \frac{d[\text{KCl}]}{dt}$$

$$\text{or } \frac{k_1}{4} = \frac{k_2}{3} = k_3$$

$$\text{or } 3k_1 = 4k_2 = 12k_3$$

Q81 Text Solution:

(2)

A process that leads to decrease in free energy will be spontaneous. Other three statements are true.

Q82 Text Solution:

(3)

Hybridisation and shape of XeF_2 molecule is sp^3d and linear.

Q83 Text Solution:

(4)

$$\therefore K = Ae^{-E_a/RT}$$

$$\text{i.e. } \ln K = \ln A - \frac{E_a}{R} \times \frac{1}{T}$$

Q84 Text Solution:

(1)

1 mol (16 g) of CH_4 produces 890.3 kJ energy

$$\text{Thus, 1 g will produce} = \frac{890.3}{16} = 55.6 \text{ kJ}$$

1 mol (58 g) of C_4H_{10} produces energy = 2878.7 kJ

$$\text{Thus, 1 g will produce} = \frac{2878.7}{58} = 49.6 \text{ kJ}$$

As 1 g of methane produces more energy than 1 g of butane thus methane is more efficient fuel.

Q85 Text Solution:

(2)

At A : temperature = 300 K, volume = 10 L

Pressure = P_1

At C : temperature = 600 K, volume = 20 L,

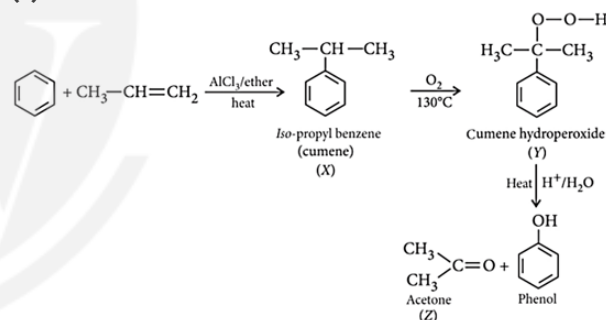
Pressure = P_2

$$\therefore \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}, \frac{P_1 \times 10}{300} = \frac{P_2 \times 20}{600}, P_1 = P_2$$

Thus, the process is isobaric.

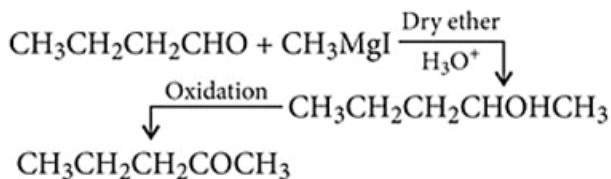
Q86 Text Solution:

(1)



Q87 Text Solution:

(2)



Q88 Text Solution:

(4)

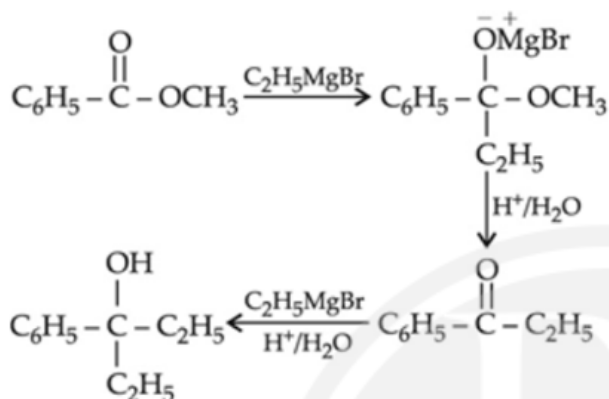


A-II, B-III, C-IV, D-I

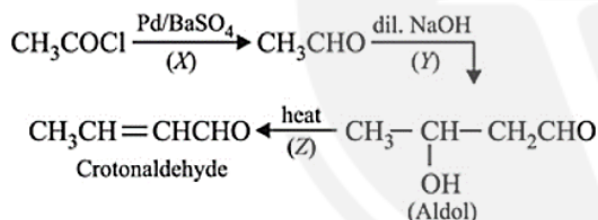
$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$	$\Delta H = \Delta U$
$N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$	$\Delta H = \Delta U - 2RT$
$NH_4HS_{(s)} \rightarrow NH_{3(g)} + H_2S_{(g)}$	$\Delta H = \Delta U + 2RT$
$PCl_{5(g)} \rightarrow PCl_{3(g)} + Cl_{2(g)}$	$\Delta H = \Delta U + RT$

Q89 Text Solution:

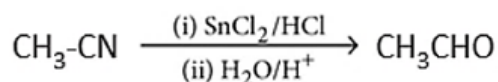
(3)

**Q90 Text Solution:**

(1)

**Q91 Text Solution:**

(2)

**Q92 Text Solution:**

(2)

$$\log k = \log A - \frac{E_a}{2.303 RT} \quad \dots \text{(i)}$$

$$\log k = 14.34 - \frac{12.5 \times 10^4 \text{K}}{T} \quad \dots \text{(ii)}$$

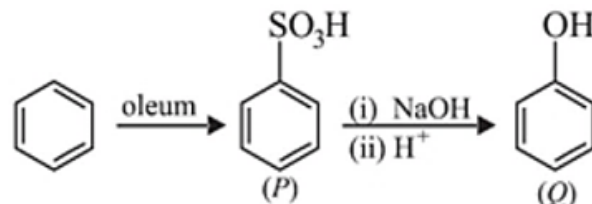
From eqs. (i) and (ii)

$$\frac{E_a}{2.303 R} = 12.5 \times 10^4 \text{K}$$

$$\begin{aligned}
 E_a &= (12.5 \times 10^4 \text{K}) \times 2.303 \times 8 \\
 &\quad .314 \text{ J K}^{-1} \text{ mol}^{-1} \\
 &= 239 \times 10^4 \text{ J mol}^{-1} = 2.39 \times 10^6 \text{ J mol}^{-1} \\
 E_a &= 2.39 \times 10^3 \text{ kJ mol}^{-1}
 \end{aligned}$$

Q93 Text Solution:

(2)

**Q94 Text Solution:**

(2)

$$k_1 = Ae^{-E_a/RT_1}$$

$$k_2 = Ae^{-E_a/RT_2}$$

$$\ln k_1 = \ln A - E_a/RT_1 \quad \text{(i)}$$

$$\ln k_2 = \ln A - E_a/RT_2 \quad \text{(ii)}$$

From eq. (i) and (ii), we have

$$\ln k_2 - \ln k_1 = \ln A - \frac{E_a}{RT_2} - \ln A + \frac{E_a}{RT_1}$$

$$\Rightarrow \ln \frac{k_2}{k_1} = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\Rightarrow \ln \frac{k_2}{k_1} = -\frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

Q95 Text Solution:

(1)

$$\Delta H_{\text{hydration}} : -15.90 - (2.75) = -18.65 \text{ kcal.}$$

Q96 Text Solution:

(4)

For 99% completion of the reaction

$$\text{If } a = 100; (a - x) = (100 - 99) = 1$$

$$t_{99\%} = \frac{2.303}{k} \log \frac{100}{1} = \frac{2.303}{k} \log 10^2$$

$$t_{99\%} = \frac{2.303 \times 2}{k} \Rightarrow t_{99\%} = \frac{4.606}{k} \quad \text{(i)}$$

For 90% completion of the reaction:

$$\text{If } a = 100; (a - x) = (100 - 90) = 10$$

[Android App](#)[iOS App](#)[PW Website](#)

$$t_{90\%} = \frac{2.303 \times 2}{k} \log \frac{100}{10} = \frac{2.303}{k} \log 10 \quad (\text{ii})$$

$$\Rightarrow t_{90\%} = \frac{2.303}{k}$$

On dividing equation (i) by equation (ii), we get

$$\frac{t_{99\%}}{t_{90\%}} = \frac{4.603}{k} \times \frac{k}{2.303} = 2$$

It means that time required for 99% completion of the reaction is twice the time required to complete 90% of the reaction.

Q97 Text Solution:

(3)

250 mL of 4 N NaOH = 500 mL of 2 N NaOH.

This will completely neutralise 500 mL of 2 N HCl.

Heat evolved from 1000 mL of 1 N HCl = x (given)

$\therefore$ Heat evolved from 500 mL of 2 N HCl = x.

Q98 Text Solution:

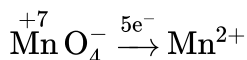
(4)

Zero, in all the cases because internal energy is a state function. For cyclic process, $\Delta U = 0$.

Q99 Text Solution:

(4)

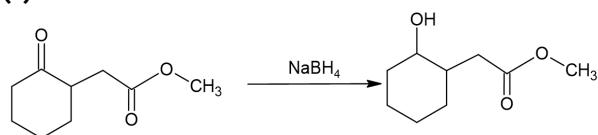
In acidic medium



$$\text{E. W.} = \frac{\text{M.W.}}{5} = \frac{158}{5} = 31.6$$

Q100 Text Solution:

(1)



Q101 Text Solution:

(2)

The pollen and stigma of different flowers of the same plant are involved in geitonogamy. Anthers

and stigma have been disclosed in chasmogamous flowers. Cleistogamous flowers are always autogamous.

[New NCERT Class 12th Page No. 12]

Q102 Text Solution:

(4)

Penicillin is formed by fungus (*Penicillium notatum*) Rum is formed by yeast.

[New NCERT Class 12th Page No. 153]

Q103 Text Solution:

(2)

	I ^A I ^B	I ^A i
I ^A	I ^A I ^A	I ^A i
i	I ^A i	ii

Number of genotypes = 4

Number of phenotypes = 3

I^AI^A and I^Ai = A

I^AI^B = AB

I^Bi = B

[New NCERT Class 12th Page No. 61]

Q104 Text Solution:

(4)

Some plants such as *Viola* (common pansy), *Oxalis*, and *Commelina* produce two types of flowers –chasmogamous flowers which are similar to flowers of other species with exposed anthers and stigma, and cleistogamous flowers which do not open at all. In such flowers, the anthers and stigma lie close to each other. When anthers dehisce in the flower buds, pollen grains come in contact with the stigma to effect pollination. Thus, cleistogamous flowers are invariably autogamous as there is no chance of cross-pollen landing on the stigma.

[New NCERT Class 12th Page No. 12]



Q105 Text Solution:

(3)

The primary effluent is passed into large aeration tanks where it is constantly agitated mechanically and air is pumped into it. This allows vigorous growth of useful aerobic microbes into flocs (masses of bacteria associated with fungal filaments to form mesh like structures).

[New NCERT Class 12th Page No. 154]**Q106 Text Solution:**

(4)

Aspergillus niger - citric acid

Trichoderma polysporum - Immunosuppressive agent

Saccharomyces cerevisiae - Ethanol

[New NCERT Class 12th Page No. 153]**Q107 Text Solution:**

(4)

- Resistance against root-borne pathogens.
- Tolerance to salinity and drought.
- Helps in absorption of phosphorous.
- Overall increase in the plant growth and development.

[New NCERT Class 12th Page No. 158]**Q108 Text Solution:**

(4)

Tapetum → Middle layers → Endothecium → Epidermis

[New NCERT Class 12th Page No. 5]**Q109 Text Solution:**

(3)

Sickle cell anaemia is qualitative disorder.

[New NCERT Class 12th Page No. 75]**Q110 Text Solution:**

(2)

Methanogens	<i>Methanobacterium</i>
-------------	-------------------------

Organic wastes	Biochemical oxygen demand
<i>Bacillus thuringiensis</i>	Microbial biocontrol agent
<i>Penicillium notatum</i>	Penicillin

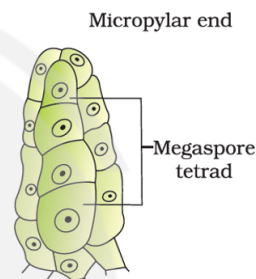
[New NCERT Class 12th Page No. 152, 154]**Q111 Text Solution:**

(3)

Khadi and Village Industries Commission

[New NCERT Class 12th Page No. 156]**Q112 Text Solution:**

(3)

[New NCERT Class 12th Page No. 10]**Q113 Text Solution:**

(4)

The number of ovules in orchids is many.

Papaver has multicarpellary syncarpous pistil.

[New NCERT Class 12th Page No. 9]**Q114 Text Solution:**

(4)

The man is suffering from Klinefelter's syndrome.

[New NCERT Class 12th Page No. 76]**Q115 Text Solution:**

(4)

It was proposed independently by Sutton and Boveri.

It shows parallelism between behaviour of chromosomes and behaviour of genes.

It was experimentally verified by T.H. Morgan.

[New NCERT Class 12th Page No. 67]

Q116 Text Solution:

(4)

As both types of gametes are non-motile, they have to be brought together for fertilisation to occur.

Depending on the source of pollen, pollination can be divided into three types.

[New NCERT Class 12th Page No. 11]

Q117 Text Solution:

(4)

Rosaceae - Viability of pollen is for months.

[New NCERT Class 12th Page No. 8]

Q118 Text Solution:

(3)

- Organic farmers work to create a system where pests are kept at manageable level not completely eradicated.
- Key belief of organic farmers is that biodiversity furthers health.

[New NCERT Class 12th Page No. 156]

Q119 Text Solution:

(3)

Brewer's yeast is used in production of ethanol.

[New NCERT Class 12th Page No. 152]

Q120 Text Solution:

(3)

Phenotypic ratio of monohybrid cross in F_2 is 3:1.

Genotypic ratio of monohybrid cross in F_2 is 1:2:1.

[New NCERT Class 12th Page No. 57]

Q121 Text Solution:

(4)

Antibiotics have greatly improved our capacity to treat deadly disease such as plague, whooping cough (kali khansi), diphtheria (gal ghotu) and leprosy (kusht rog), which used to kill millions all over the globe.

[New NCERT Class 12th Page No. 153]

Q122 Text Solution:

(3)

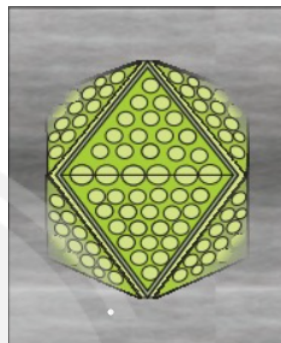
BOD is the amount of oxygen that would be consumed if all the organic matter in one litre of water were oxidized by bacteria.

[New NCERT Class 12th Page No. 157]

Q123 Text Solution:

(2)

The given virus is Adenovirus.



[New NCERT Class 12th Page No. 150]

Q124 Text Solution:

(3)

Microsporogenesis occurs in sporogenous tissue through meiotic division.

[New NCERT Class 12th Page No. 6]

Q125 Text Solution:

(1)

The people who have undergone myocardial infarction are given streptokinase for the removal of clots from the blood vessels.

[New NCERT Class 12th Page No. 153]

Q126 Text Solution:

(4)

In western countries, a large number of pollen products in the form of tablets and syrups are available in the market. It is possible to store pollen grains of a large number of species for years in liquid nitrogen.



[New NCERT Class 12th Page No.8]**Q127 Text Solution:**

(1)

Integument	One or two protective layers of ovule
Hilum	Region where body of ovule fuses with funicle
Funicle	Stalk of ovule
Chalaza	Basal part of ovule

[New NCERT Class 12th Page No. 9]**Q128 Text Solution:**

(4)

Curd increases B₁₂ vitamin and prevents the increase of disease causing microbes.

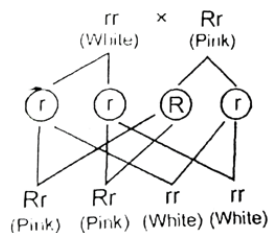
[New NCERT Class 12th Page No. 151]**Q129 Text Solution:**

(2)

Although geitonogamy is functionally cross-pollination involving a pollinating agent, genetically it is similar to autogamy since the pollen grains come from the same plant.

[New NCERT Class 12th Page No.12]**Q130 Text Solution:**

(4)



Phenotypic ratio

Pink : White

1 : 1

Genotypic ratio

Rr : rr

1 : 1

[New NCERT Class 12th Page No. 60]**Q131 Text Solution:**

(2)

Queen victoria shows a number of haemophilic descendent as she was normal but carrier.

[New NCERT Class 12th Page No. 74]**Q132 Text Solution:**

(3)

- Blue green algae serves as an important biofertiliser in paddy fields.
- Blue green algae can fix atmospheric nitrogen as well as adds organic matter to soil.

[New NCERT Class 12th Page No. 158]**Q133 Text Solution:**

(1)

Beta thalassemia is controlled by single gene HBB on chromosome 11.

[New NCERT Class 12th Page No. 75]**Q134 Text Solution:**

(4)

Biogas includes gases like methane, CO₂ & H₂S.

[New NCERT Class 12th Page No. 154]**Q135 Text Solution:**

(3)

Mendel selected 14 true breeding pea plant varieties.

[New NCERT Class 12th Page No. 54]**Q136 Text Solution:**

(4)

Wine and beer without distillation. Whisky, brandy and rum with distillation.

[New NCERT Class 12th Page No. 152]**Q137 Text Solution:**

(3)

Water lily and water hyacinth are insect or wind pollinated.

[New NCERT Class 12th Page No. 13]**Q138 Text Solution:**[Android App](#)[iOS App](#)[PW Website](#)

(4)

Colour blindness is carried on the X-chromosome and can only be inherited by the son if the mother is a carrier.

[New NCERT Class 12th Page No. 73]

Q139 Text Solution:

(4)

- Human males = XY
- Female hen = ZW
- Male birds = ZZ
- Male grasshopper = XO

[New NCERT Class 12th Page No. 73]

Q140 Text Solution:

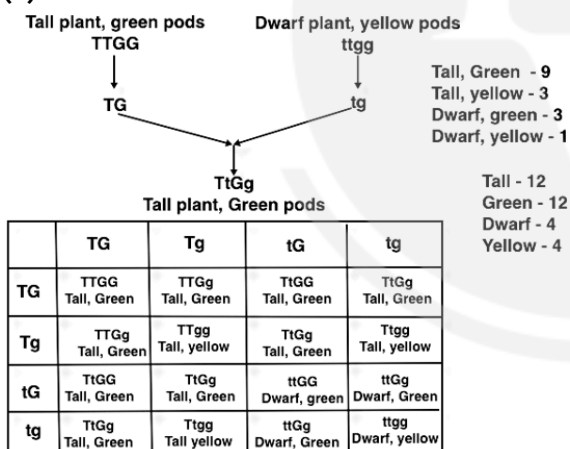
(1)

Segregation of mendelian factors occurs during anaphase-I.

[New NCERT Class 12th Page No. 66]

Q141 Text Solution:

(3)



[New NCERT Class 12th Page No. 33]

Q142 Text Solution:

(3)

A-Plumule B-Radicle

[New NCERT Class 12th Page No.19]

Q143 Text Solution:

(4)

Alfred Sturtevant	Mapped position of genes
Henking	X-body
Correns	Involved in rediscovery of Mendel's law on the inheritance of characters
Morgan	Linkage

[New NCERT Class 12th Page No. 65, 67, 69]

Q144 Text Solution:

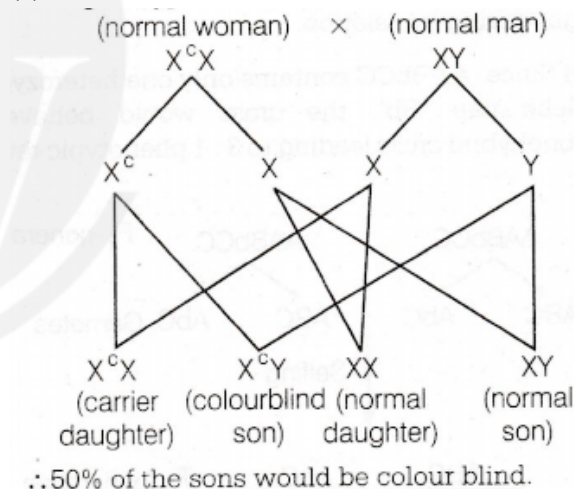
(1)

According to these plans, treated sewage can be discharged into these rivers.

[New NCERT Class 12th Page No. 155]

Q145 Text Solution:

(1)



[New NCERT Class 12th Page No. 73]

Q146 Text Solution:

(3)

One of the male gametes moves towards the egg cell and fuses with its nucleus thus completing the syngamy. This results in the formation of a diploid cell, the zygote. The other male gamete



moves towards the two polar nuclei located in the central cell and fuses with them to produce a triploid primary endosperm nucleus.

[New NCERT Class 12th Page No. 18]

Q147 Text Solution:

(2)

Morgan carried out several dihybrid cross to study genes that were showing independent assortment dihybrid.

[New NCERT Class 12th Page No. 67]

Q148 Text Solution:

(2)

The coconut water from tender coconut that you are familiar with, is nothing but free-nuclear endosperm (made up of thousands of nuclei) and the surrounding white kernel is the cellular endosperm.

[New NCERT Class 12th Page No. 18, 19]

Q149 Text Solution:

(4)

Cashew is false fruit.

[New NCERT Class 12th Page No. 20]

Q150 Text Solution:

(1)

- Selection of parents
- Emasculation
- Bagging
- Dusting the pollen on stigma
- Re-bagging

[New NCERT Class 12th Page No.17]

Q151 Text Solution:

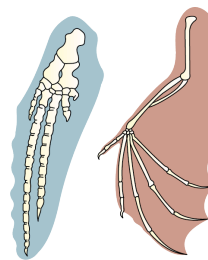
(1)

- Hepatitis-B is considered as sexually transmitted disease.
- It can be spread through unprotected sexual contact.

[New NCERT Class 12th Page No. 47]

Q152 Text Solution:

(3)



- X and Y appear to represent homologous organs.
- Homologous organs are those that have the same basic structure and developmental origin but may perform different functions. For example, the forelimbs of vertebrates (like the structure in the diagrams) serve different functions (e.g., flight in birds and grasping in mammals) but share a common anatomical pattern.

[New NCERT Class 12th Page No. 112]

Q153 Text Solution:

(3)

The sperm head contains an elongated haploid nucleus, the anterior portion of which is covered by a cap-like structure, acrosome.

[New NCERT Class 12th Page No. 32]

Q154 Text Solution:

(4)

About 15 mya, primates called *Dryopithecus* and *Ramapithecus* were existing. They were hairy and walked like gorillas and chimpanzees.

[New NCERT Class 12th Page No. 124]

Q155 Text Solution:

(3)

- In disruptive natural selection, more individuals acquire peripheral character value at both ends of the distribution curve.
- Cave paintings by Pre-historic humans can be seen at Bhimbetka rock shelter.



[New NCERT Class 12th Page No. 120 and 124]

Q156 Text Solution:

(1)

- Vas deferens receives a duct from seminal vesicle and opens into urethra as the ejaculatory duct.
- Parts of the female reproductive system along with a pair of the mammary glands are integrated structurally and functionally to support the processes of ovulation, fertilisation, pregnancy, birth and child care.

[New NCERT Class 12th Page No. 28]

Q157 Text Solution:

(3)

Hormone releasing IUDs	Progestasert
Copper releasing IUDs	Multiload 375
Non-medicated IUDs	Lippes loop
Barrier contraceptive method	Cervical caps

[New NCERT Class 12th Page No. 44]

Q158 Text Solution:

(2)

Examples of organisms which have evolved due to changes in environment brought about by anthropogenic action are:

- Herbicides resistant weeds.
- Drug resistant bacteria.

[New NCERT Class 12th Page No. 116]

Q159 Text Solution:

(3)

Evolution is a stochastic process. It is based on chance events in nature and chance mutation in the organisms.

[New NCERT Class 12th Page No. 116]

Q160 Text Solution:

(2)

Secretions of the male accessory glands—seminal vesicle, prostate and bulbourethral gland constitute the seminal plasma which is rich in fructose, calcium and certain enzymes.

[New NCERT Class 12th Page No. 28]

Q161 Text Solution:

(4)

- MTPs are safe during the first trimester.
- Government of India legalised MTP in 1971 with some strict conditions to avoid its misuse.

[New NCERT Class 12th Page No. 46]

Q162 Text Solution:

(4)

A primary spermatocyte completes the first meiotic division (reduction division) leading to formation of two equal, haploid cells called secondary spermatocytes, which have only 23 chromosomes each.

[New NCERT Class 12th Page No. 31]

Q163 Text Solution:

(1)

According to Hardy Weinberg equilibrium, allele frequencies in a population are stable and is constant from generation to generation.

[New NCERT Class 12th Page No. 120]

Q164 Text Solution:

(2)

The formation of life was preceded by chemical evolution, i.e., the formation of diverse organic molecules from inorganic constituents.

[New NCERT Class 12th Page No. 111 and 112]

Q165 Text Solution:

(4)

- Spermatogonia and primary spermatocytes are the diploid cells while secondary spermatocytes and spermatids along with spermatozoa are haploid cells.



- If $2n = 16$ is in a primary spermatocyte, then the haploid number would be 8. Hence, 8 chromosomes will be present in each secondary spermatocyte.

[New NCERT Class 12th Page No. 31]

Q166 Text Solution:
(3)

- Brain capacity of *Homo habilis* is 650-800cc.
- Brain capacity of *Homo erectus* is 900cc.
- Brain capacity of Neanderthal man is 1400cc..

[New NCERT Class 12th Page No. 124, 125]

Q167 Text Solution:
(4)

- Branching descent and natural selection are the two key concepts of Darwinian Theory of Evolution.
- Fitness is the end result of the ability to adapt and get selected by nature.

[New NCERT Class 12th Page No.118]

Q168 Text Solution:
(2)

All these characteristics shown by Neanderthal man.

[New NCERT Class 12th Page No. 124]

Q169 Text Solution:
(1)

- The glandular tissue of each breast is divided into 15-20 mammary lobes containing clusters of cells called alveoli.
- The alveoli open into mammary tubules.
- The tubules of each lobe join to form a mammary duct.
- Several mammary ducts join to form a wider mammary ampulla which is connected to lactiferous duct through which milk is sucked out.
- The correct sequence of transport of milk

is; Mammary tubule → Mammary duct → Mammary ampulla → Lactiferous duct.

[New NCERT Class 12th Page No. 31]

Q170 Text Solution:
(3)

According to the theory of panspermia, units of life called spores were transferred to different planets including Earth.

[New NCERT Class 12th Page No. 111]

Q171 Text Solution:
(2)

- As the name implies, Ichthyosaurs are 200 million years old reptiles resembling fish. Their forebears were land-based reptiles that later returned to the water and transformed into fish-like reptiles, now extinct.
- Dinosaurs became extinct at the end of the Cretaceous Period, about 65 million years ago.

[New NCERT Class 12th Page No. 122,123]

Q172 Text Solution:
(4)

In 1953, S. L. Miller created the primitive earth conditions in the laboratory and gave experimental evidence for the origin of first form of life from pre-existing non-living organic molecules. The primitive earth conditions created include high temperature, volcanic storms, reducing atmosphere.

[New NCERT Class 12th Page No. 111]

Q173 Text Solution:
(3)

Implants are used by females only.

[New NCERT Class 12th Page No. 44]

Q174 Text Solution:
(3)

- Spermicidal creams, jellies and foams are usually used along with barriers to increase



their contraceptive efficiency.

- Genital herpes is not curable even if it is detected early and treated properly.

[New NCERT Class 12th Page No. 44 and 47]

Q175 Text Solution:

(3)

Bobcat is similar to Tasmanian tiger cat in the same way Lemur is similar to spotted cuscus.

[New NCERT Class 12th Page No. 118]

Q176 Text Solution:

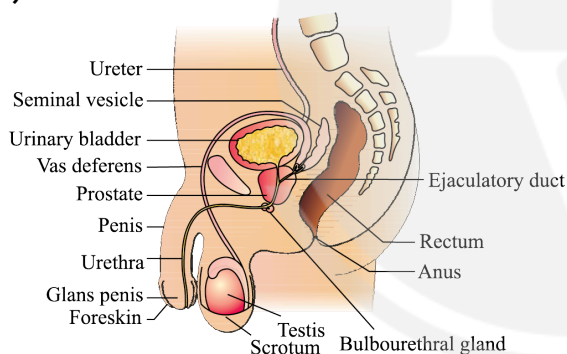
(3)

During post-industrialisation period, the tree trunks became dark due to industrial smoke and soots. Under this condition the white-winged moth did not survive due to predators whereas dark winged or melanised moth survived.

[New NCERT Class 12th Page No. 115]

Q177 Text Solution:

(4)



Vas deferens is a duct that ascends to the abdomen and loops over the urinary bladder.

[New NCERT Class 12th Page No. 27]

Q178 Text Solution:

(1)

The chorionic villi and uterine tissue become interdigitated with each other and jointly form a structural and functional unit between developing embryo (foetus) and maternal body called placenta.

[New NCERT Class 12th Page No. 37]

Q179 Text Solution:

(3)

Oxytocin acts on the uterine muscle and causes stronger uterine contractions, which in turn stimulates further secretion of oxytocin. This leads to expulsion of the baby out of the uterus through the birth canal – parturition.

[New NCERT Class 12th Page No.38]

Q180 Text Solution:

(3)

- Zygote intrafallopian transfer is an *in-vitro* fertilisation technique.
- The menstrual flow occurs due to breakdown of endometrial lining of the uterus and its blood vessels which forms liquid that comes out through vagina.
- Menstruation only occurs if the released ovum is not fertilised.

[New NCERT Class 12th Page No. 34,48]

Q181 Text Solution:

(3)

Placenta acts as an endocrine tissue and produces several hormones like human chorionic gonadotropin (hCG), human placental lactogen (hPL), estrogens, progesterones, etc.

[New NCERT Class 12th Page No. 37]

Q182 Text Solution:

(3)

- By the time of 500 mya, invertebrates were formed and active.
- Earth is about 4000 years old according to special creation theory.

[New NCERT Class 12th Page No. 112 and 122]

Q183 Text Solution:

(4)

Mole	Marsupial mole
------	----------------



<i>Homo erectus</i>	Fossils discovered in Java.
Neanderthal man	Lived in between 10,000-40,000 years back.
Anteater	Numbat

[New NCERT Class 12th Page No. 118, 125]

Q184 Text Solution:

(1)

- Copper releasing IUDs increase phagocytosis of sperms within the uterus.
- Cu ions released suppress sperm motility and their fertilising capacity.

[New NCERT Class 12th Page No. 44]

Q185 Text Solution:

(4)

The frequency of dominant allele = 0.7

From Hardy Weinberg principle, $P + q = 1$

Frequency of recessive allele (q) = 0.3

So the frequency of heterozygotes = $2pq = 0.42$

[New NCERT Class 12th Page No. 121]

Q186 Text Solution:

(3)

The cavity of the cervix is called cervical canal which along with vagina forms the birth canal.

[New NCERT Class 12th Page No. 30]

Q187 Text Solution:

(4)

Condom	Barrier contraceptive method
Saheli	Non-steroidal pill
LNG-20	Hormone releasing IUD
Withdrawal method	Natural contraceptive method

[New NCERT Class 12th Page No. 44 and 45]

Q188 Text Solution:

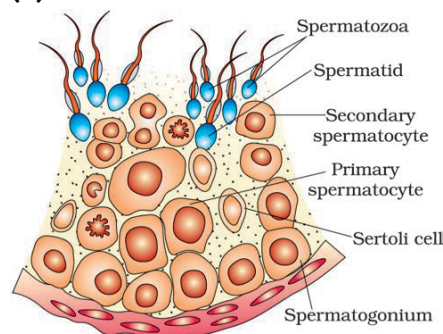
(4)

Gonorrhoea is a venereal disease that are transmitted through sexual intercourse.

[New NCERT Class 12th Page No. 35, 37, 45 and 47]

Q189 Text Solution:

(2)



- Diagrammatic sectional view of a seminiferous tubule (enlarged)
- The spermatids are transformed into spermatozoa (sperms) by the process called spermiogenesis.

[New NCERT Class 12th Page no.- 31]

Q190 Text Solution:

(1)

'Nirodh' is a popular brand of condom for the male.

[New NCERT Class 12th Page No. 44 and 47]

Q191 Text Solution:

(1)

Diaphragms, cervical caps and vaults are also barriers made of rubber that are inserted into the female reproductive tract to cover the cervix during coitus. They prevent conception by blocking the entry of sperms through the cervix. They are reusable. Diaphragms, cervical caps and vaults are also barriers made of rubber that are inserted into the female reproductive tract to cover the cervix during coitus. They prevent conception by blocking the entry of



sperms through the cervix.

They are reusable.

[New NCERT Class 12th Page No. 44]

Q192 Text Solution:

(1)

In the later phase of pregnancy, a hormone called relaxin is also secreted by the ovary.

[New NCERT Class 12th Page No. 36, 37 and 38]

Q193 Text Solution:

(1)

Following is the correct sequence of evolution of plant forms;

Chlorophyte ancestors → Tracheophyte ancestors → Rhynia type plants → Psilophyton.

[New NCERT Class 12th Page No. 138]

Q194 Text Solution:

(3)

The process of formation of a mature female gamete is called oogenesis which is markedly different from spermatogenesis. Oogenesis is initiated during the embryonic development stage when a couple of million gamete mother cells (oogonia) are formed within each fetal ovary; no more oogonia are formed and added after birth.

[New NCERT Class 12th Page No. 32 and 33]

Q195 Text Solution:

(2)

Fossils	Palaeontological evidence
Theory of abiogenesis	Living organisms originated from non-living matter
Convergent evolution	Analogous organs

<i>Homo habilis</i>	First human-like being
---------------------	------------------------

[New NCERT Class 12th Page No. 111]

Q196 Text Solution:

(3)

Primary oocyte within the tertiary follicle grows in size and completes its first meiotic division.

[New NCERT Class 12th Page No. 27, 29 and 32]

Q197 Text Solution:

(1)

- Contraceptives are not regular requirements for the maintenance of reproductive health.
- Amniocentesis diagnose genetic disorders.

[New NCERT Class 12th Page No. 42 and 44]

Q198 Text Solution:

(3)

In tubectomy, a small part of the fallopian tube is removed or tied up through a small incision in the abdomen or through the vagina.

[New NCERT Class 12th Page No. 45]

Q199 Text Solution:

(4)

Natural selection acts on existing variation but does not create it.

[New NCERT Class 12th Page No. 132]

Q200 Text Solution:

(4)

- Adaptive radiation is the evolutionary process in which many species originate from one species in an area and radiate to different places.
- Darwin's finches represent one of the best examples of adaptive radiation.

[New NCERT Class 12th Page No. 116]



[Android App](#)

| [iOS App](#)

| [PW Website](#)